



MACHINE TRANSLATION

PROJECT GOAL

- Build an English-to-Arabic translation model.
- Start from a pretrained Transformer and fine-tune it on parallel text.
- Evaluate quality using BLEU and show real translation examples.



TOOLS

- Hugging Face: Transformers, Datasets, Evaluate
- Metric: SacreBLEU
- Demo UI: Gradio



HUGGING FACE

DATASET

Dataset Used:

- OPUS-100 English/Arabic parallel corpus
- Source: MagedSaeed/opus-100_ar_en_experimental

Split Strategy:

- Train set split into 90% train / 10% validation
- Separate test set used for final evaluation



DATA PREPROCESSING

Tokenization Pipeline

- Input: English text
- Target: Arabic text
- Uses `text_target=` for modern seq2seq tokenization

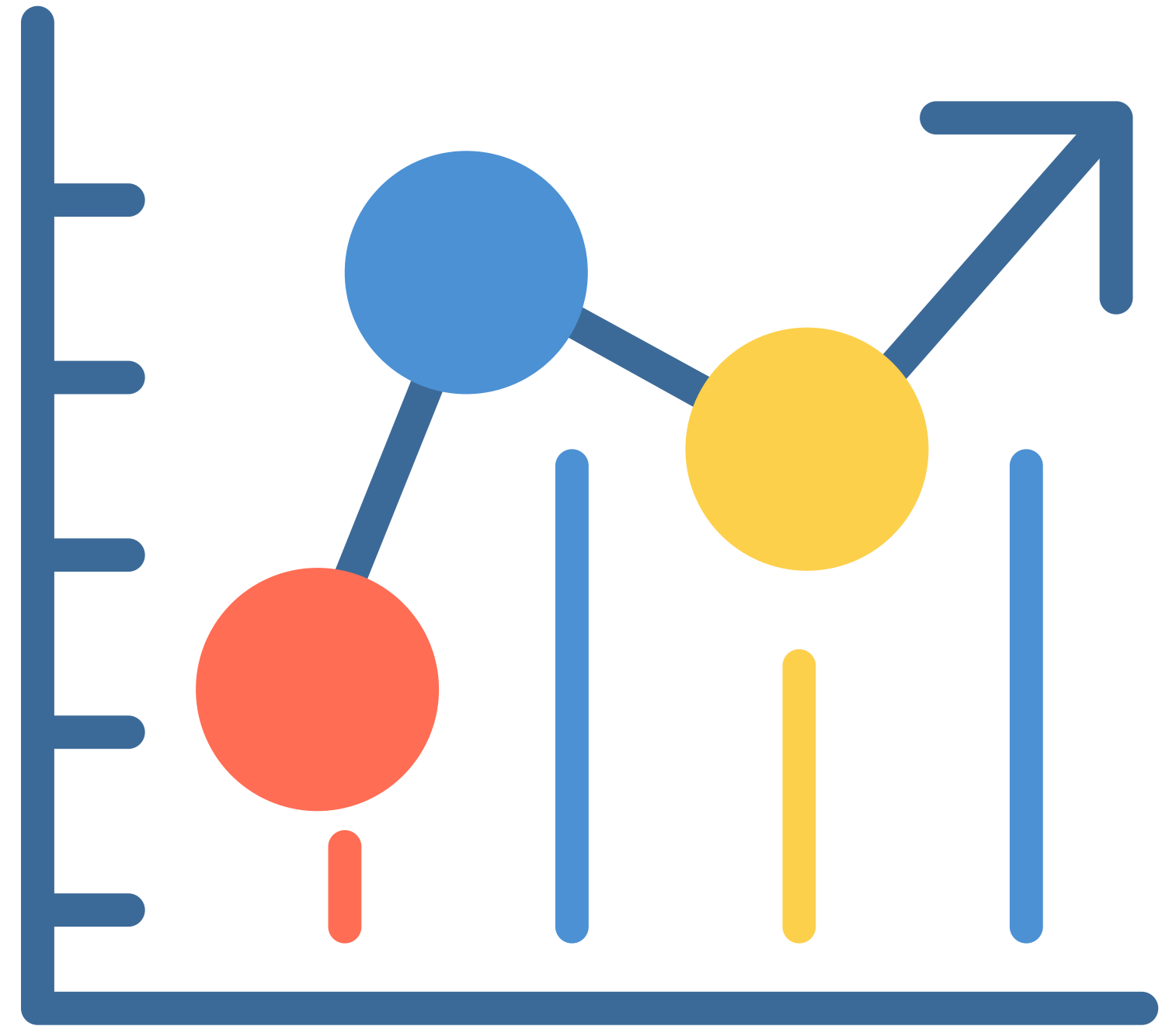
Output

- Tokenized train, validation, and test datasets

TRAINING CONFIGURATION

Trainer Setup (Seq2SeqTrainer)

- Epochs: 3
- Learning rate: $1e-5$
- Batch size: 16 (train) / 16 (eval)
- Optimizer: Adafactor



EVALUATION

Metric: SacreBLEU

- Measures translation similarity vs reference
- Used during evaluation and for selecting best model

Two Types of Evaluation

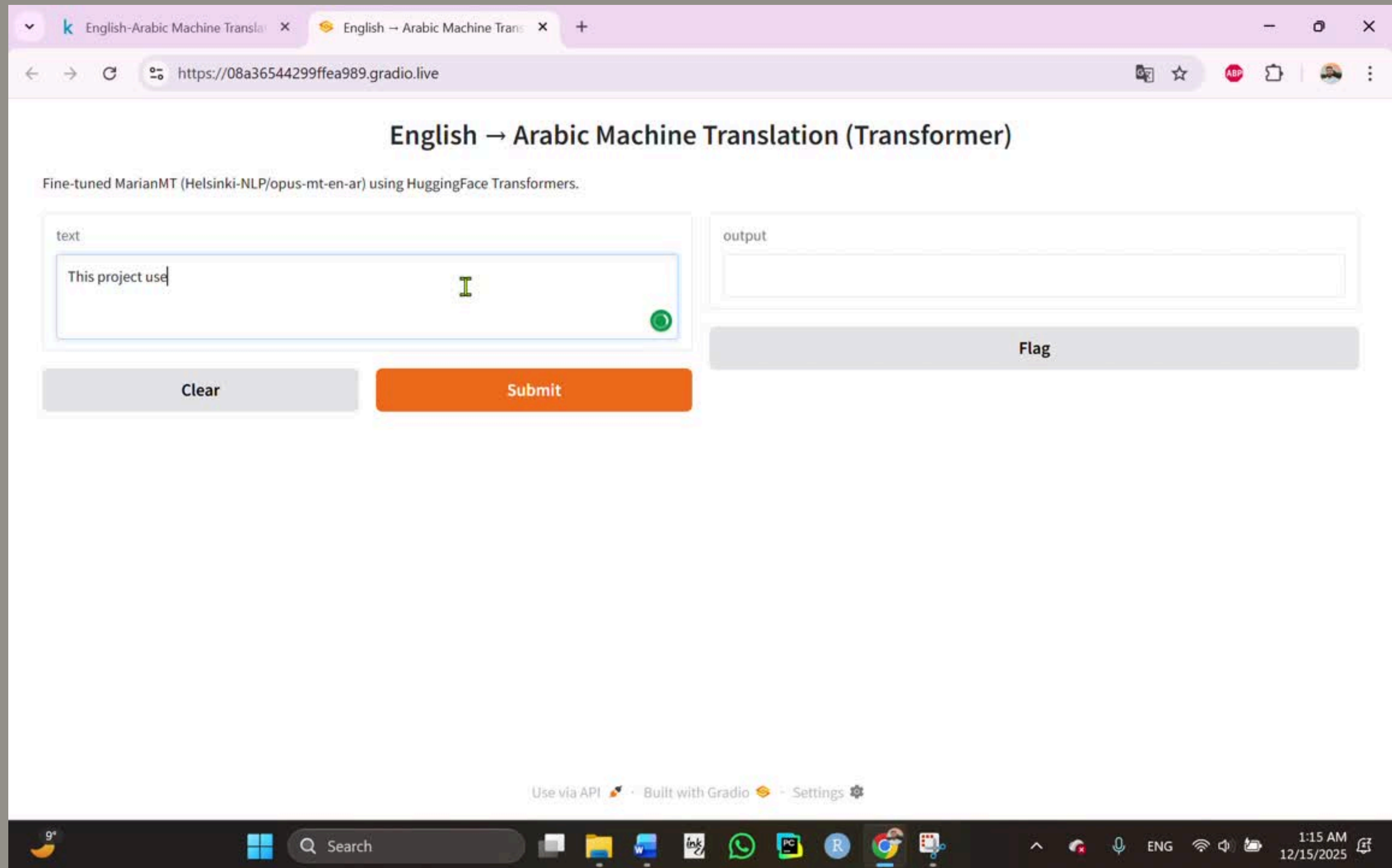
- Quantitative: BLEU score on the test set
- Qualitative: random examples (EN → AR prediction vs AR reference)



INTERACTIVE DEMO (GRADIO)

**Simple
textbox:
type English
sentence**

**Output:
Arabic
translation**



KAGGLE NOTEBOOK CODE



English-Arabic Machine Translation (Transformer)

Explore and run machine learning code with Kaggle Notebooks | Using data from No attached data sources

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THANK YOU

Presented to Dr. Suhaila



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